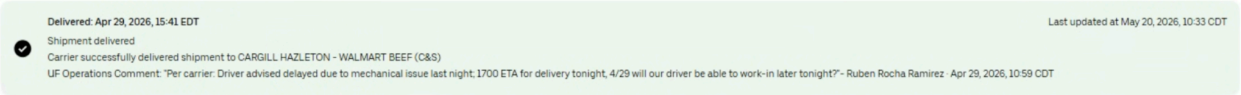
Public Tracking Portal - AI Summary

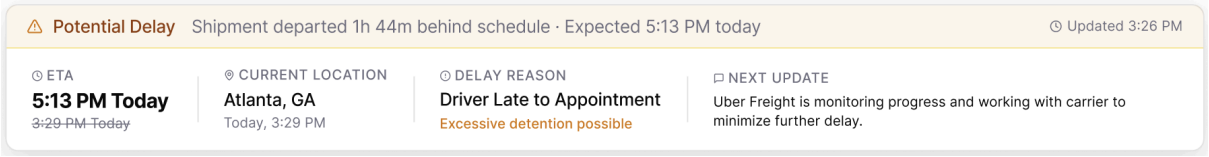
Context
Customers use the Public Tracking Portal to understand where their shipment is, when it will arrive, and whether there is a problem. Today, those answers are fragmented across the page, creating confusion and unnecessary support inquiries. The page currently feels like a shipment record, not a tracking experience. The AI Summary provides a single, trustworthy overview of shipment status, risks, and next steps at a glance.

Pain Points
- Customers must piece together what info is most relevant to shipment status
    - Across banners, ETAs, maps, comments, and route details, increasing cognitive load and making it difficult to quickly understand what is happening.
- The banner doesn't effectively answer the customers most important questions
    - Where is my shipment? When will it arrive? What changed? What happens next?
- The page does not clearly explain shipment status, ETA changes, and risk changes
    - Creating uncertainty during exceptions and unnecessary support tickets
- The page's information hierarchy is weak with many similarly weighted elements
    - Making it more challenging for customers to reconcile multiple modules
- Raw ops comments aren't revised for customer understanding

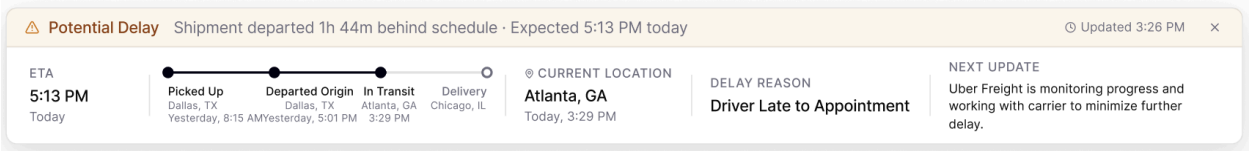
Solution
Replace existing static PTP banner with an AI shipment summary that helps customers understand the current shipment state in seconds. The summary card should go beyond displaying raw shipment events, it should interpret the most relevant customer-safe signals and present a clear, visually appealing answer to: where is my shipment? What changed? Should I be concerned? And do I need to take action?



High-Level UX Requirements
The summary will appear in the hero area of the PTP, where today's static banner appears. The default card should be visually compact, easy to scan, and contain the following information:
- Status: On Track, Potential Delay, Waiting for Update, Delivered, or Cancelled
- ETA or Delivered Time:
- Current Location:
    - If location is fresh, show current city/state
    - If location is stale, show "Last seen near [location]"
- Latest Update: AI-powered one sentence message that scans shipment data, tracking activity, ETA changes, dwell, check calls, and carrier/ops comments.
- Visual Aid Timeline
    - Horizontal journey indicator across stages of picked up, departed origin, in transit, etc



V2



Summary Content Logic

- Title: should follow priority order… if multiple are true show priority
    - Cancelled, Delivered, Delayed, Potential Delay, Waiting for Update, Carrier Not Assigned, Appointment Needs Review, In Transit
- ETA/Delivered Time
    - If shipment is in transit: show predicted ETA
    - If shipment is delivered: show delivered timestamp
    - If pickup is pending: show pickup appointment
    - If ETA is unavailable: show unavailable state
- Current Location
    - Show most recent location
        - If fresh: show city/state
        - If stale: label it as “Last seen near…”
        - If no reliable location exists: show “Location update unavailable.”
- Latest Update
    - Primary AI-generated sentence and should explain most relevant customer-facing context
    - UX Requirements
        - One sentence by default, explaining current shipment status/update
        - It should be placed just below the title and use plain customer language
        - Logic:
            - Define current shipment state
                - Delivery completed or shipment cancelled
                - Delay Reason: carrier-reported issue, missed appointment, facility dwell, stale tracking, late departure, or ETA change
                - Shipment progressing normally
                - Fallback
            - Identify highest-priority customer signal
            - Select approved message pattern
            - Fill with shipment-specific facts
            - Validate against allowed facts and desired language
- What to Expect Next
    - Should tell customer when they should expect the next meaningful update
    - UX Requirements
        - Should answer: When will I know more? Or What is UF doing to support me?
        - Field should be one concise sentence

- ** Could build logic with respect to title

Regeneration
The summary should regenerate when one of the inputs meaningfully changes.

- ETA changes meaningfully
- Shipment lifecycle status changes
- Appointment time changes
- New location update is received
- Tracking becomes stale or resumes after being stale
- Facility dwell threshold is crossed
- Carrier/ops update is received
- Risk level changes

Stop regenerating when: Shipment is delivered or cancelled, and no new customer update exist

Data & Guardrails

Data

- TMS shipment state
- Pickup and dropoff appointment times
- Unified ETA API
- Latest valid location update
- Tracking freshness
- Check call signals
- Facility dwell
- Pickup and delivery events
- Carrier-reported updates + ops comments
- Existing PTP banner logic

Guardrails
The summary must not:

- Contradict existing PTP data
- Contradict automated customer emails
- Expose internal-only comments or operational notes
- Invent delay reasons
- Blame carriers, facilities, or customers
- Present stale or low-confidence data as certain
- Claim customer action is required unless confirmed by rule based logic

If data is stale or missing, the system should use a fallback message instead of AI summary
E.g. “Waiting for an update. We do not have enough recent shipment information to generate a reliable summary.”

Goals & Success Metrics

The AI Summary should help customers understand shipment status in seconds by clearly showing what is happening, what changed, and what to expect next. The feature should reduce avoidable shipment-status outreach while maintaining trust through accurate, customer-safe summaries.

- North Star
    - Reduction in shipment-status inquires per eligible PTP Shipment view
- Customer Metrics
    - Self service completion rate: % of PTP visits with no follow up contact within 24 hours
    - Summary helpfulness rate: % of users who mark the summary as helpful
- Operational Metrics
    - Reduction in manual ops responses for eligible shipments
    - Estimated Ops time saved from avoided shipment-status inquiries
- AI Quality Metrics
    - Summary accuracy rate: % of claims supported by approved source data
    - Contradiction rate with PTP fields
    - Fallback rate when data is stale, missing, or conflicting